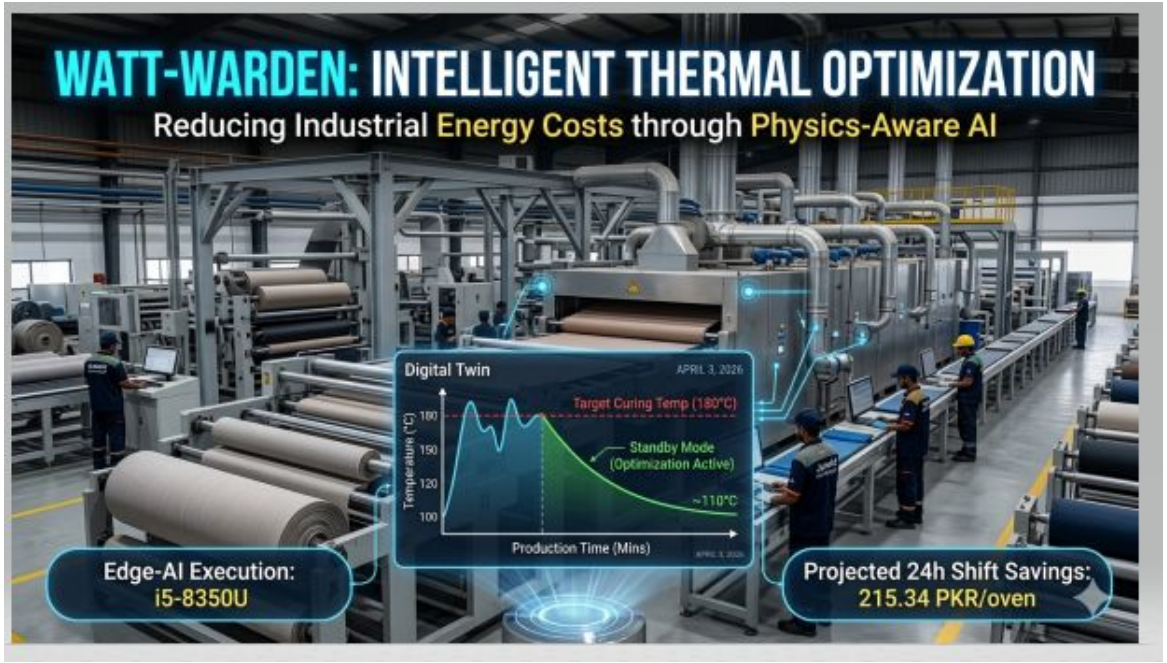
Watt-Warden: Intelligent Thermal Optimization

Reducing Industrial Energy Costs through Physics-Aware AI



The Hidden Cost of Staying "Safe"

The Reality: Industrial ovens run at 100% power during production gaps to avoid delays.

The Cost: In Karachi’s current B2 Tariff landscape, 45 minutes of "idle" time is pure profit loss.

The Risk: Manual shutdowns are slow and human error leads to cold ovens and production bottlenecks.

Watt-Warden: Your 24/7 Energy Guard

Physics-First: We don't guess. We use **Newton’s Law of Cooling** to model your specific oven's thermal DNA.

Agentic Intelligence: A three-tier AI system that:

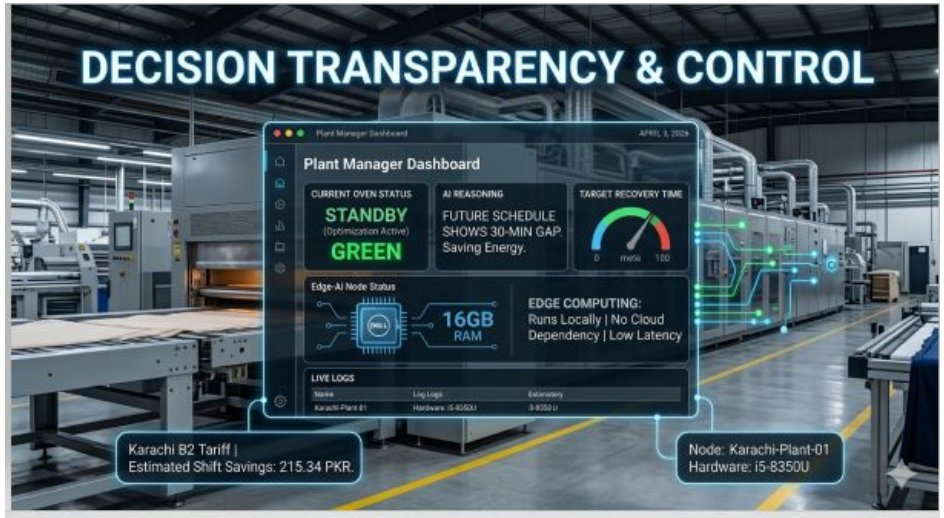
1. **Observes** real-time temperature.
2. **Plans** based on your upcoming production schedule.
3. **Executes** standby modes with surgical precision.

Decision Transparency & Control

Live Dashboard: A web-based interface for plant managers.

Smart Alerts: Clear visual indicators (Green/Red) showing the AI's logic.

Edge Computing: Runs locally on standard office hardware (i5/16GB RAM). No expensive cloud or internet dependency required.

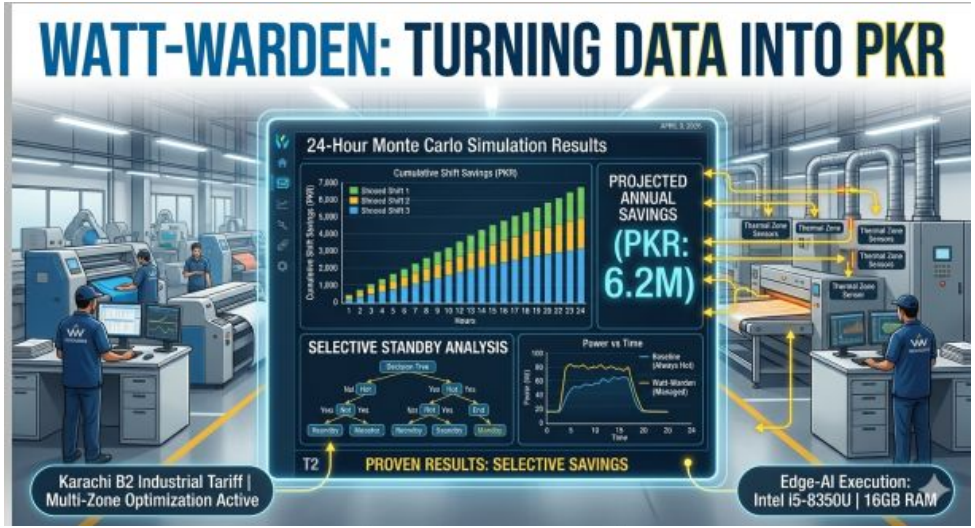


Turning Data into PKR

Proven Results: 24-hour Monte Carlo simulations show significant savings per shift.

Selective Savings: The AI only enters standby when the energy saved outweighs the cost of reheating.

Scalability: A single deployment can pay for itself within months by optimizing multiple thermal zones.

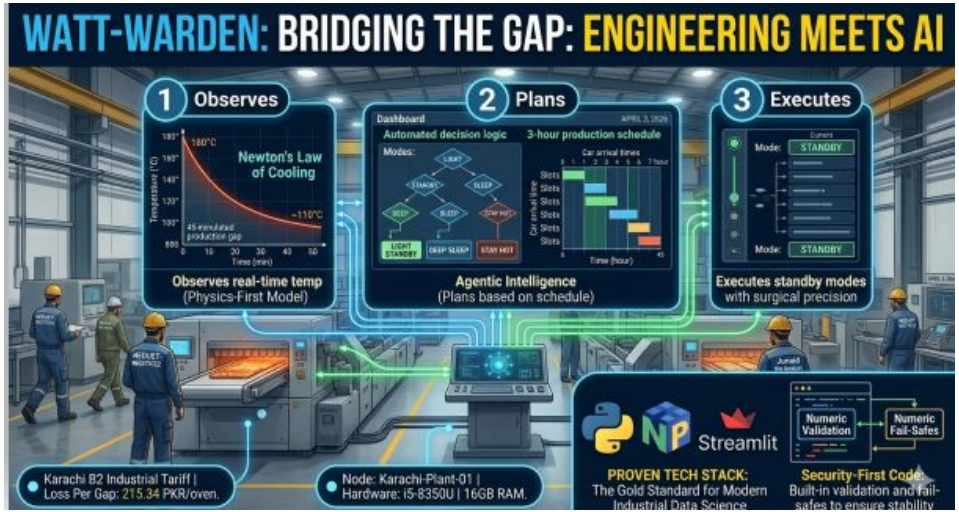


Bridging the Gap: Engineering meets AI

Domain Expertise: Developed by a NEDUET Textile Engineer with 13 years of STEM experience.

Security-First Code: Built-in numeric validation and fail-safes to ensure industrial stability.

Proven Tech Stack: Python, NumPy, and Streamlit—the gold standard for modern industrial data science.



Ready to Optimize?

The Offer: I am available to discuss:

- Custom Energy Audits for Thermal Systems.
- Digital Twin Implementation for Production Lines.
- Agentic AI Integration for Industrial Workflows.